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Sawalski et al.

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(54) **REPELLANT STRING LIGHT AND
RETROFIT ASSEMBLY**

(58) **Field of Classification Search**

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F21S 4/10; F21V 23/001; F21V 33/0092;
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(51) **Int. Cl.**
A01M 29/12 (2011.01)
F21S 4/10 (2016.01)

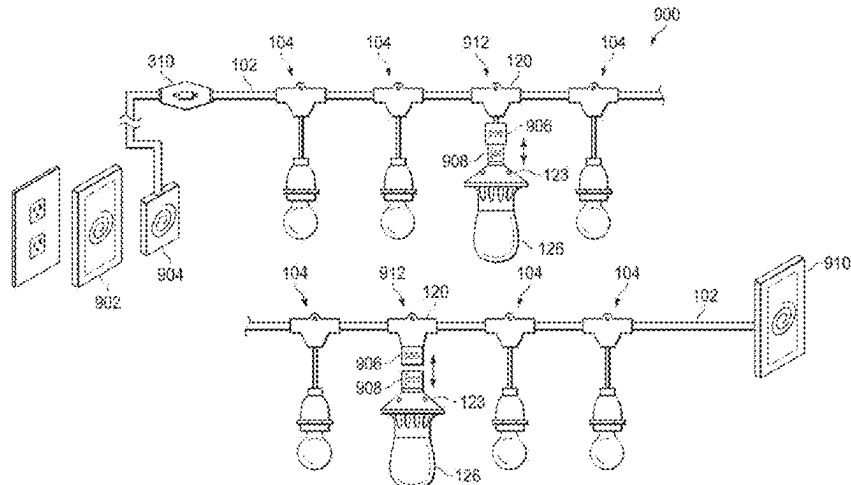
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(57) **ABSTRACT**

A repellent assembly has a light socket connector, a heating
receptacle that selectively connects to a replaceable repellent
pod that is selectively heated to disperse repellent from the
repellent pod, and a remote controlled switch interposing the
light socket connector and the heating receptacle to power
the heating receptacle on and off.

8 Claims, 19 Drawing Sheets



Related U.S. Application Data

application No. 17/101,827, filed on Nov. 23, 2020, now Pat. No. 11,754,279, which is a continuation-in-part of application No. 16/532,128, filed on Aug. 5, 2019, now Pat. No. 10,962,219.

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See application file for complete search history.

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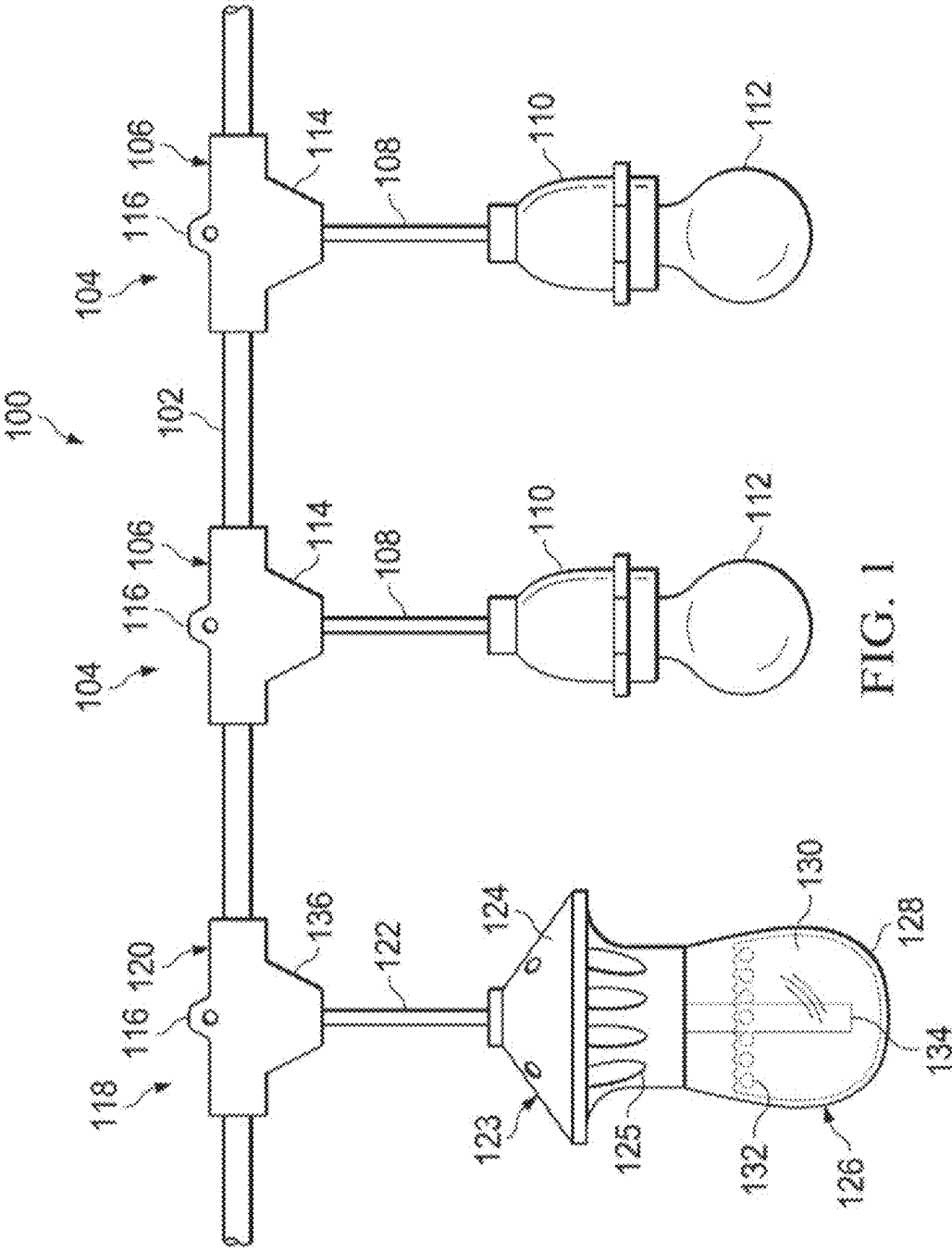
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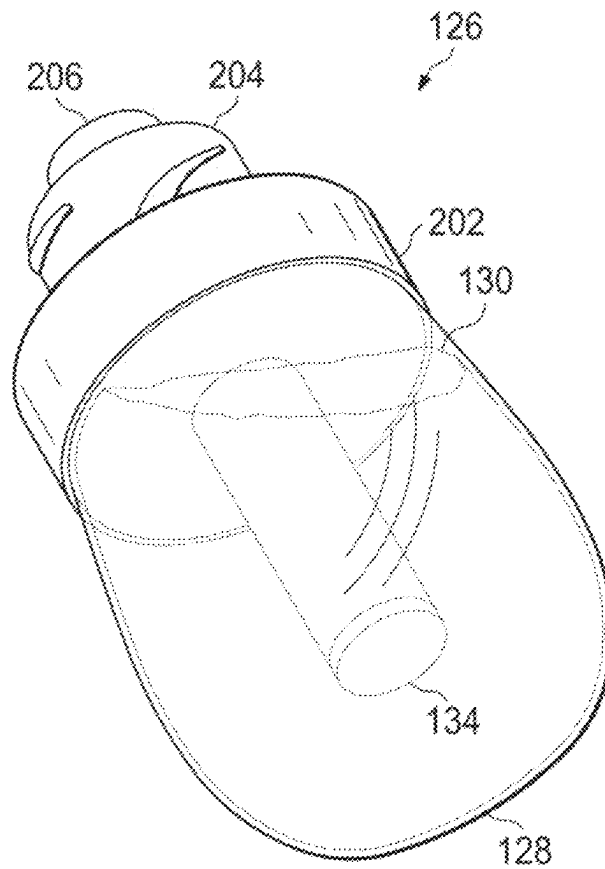
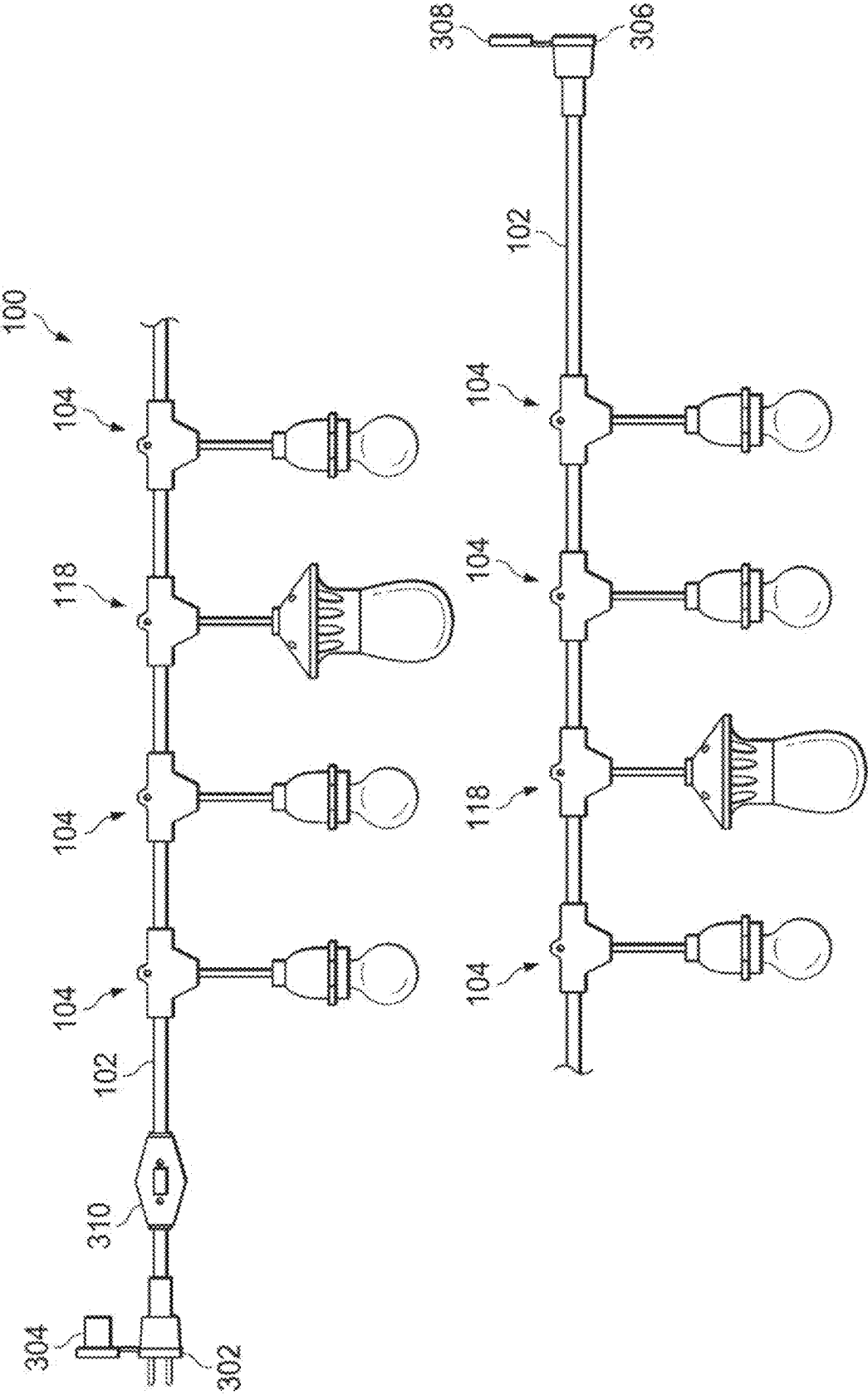
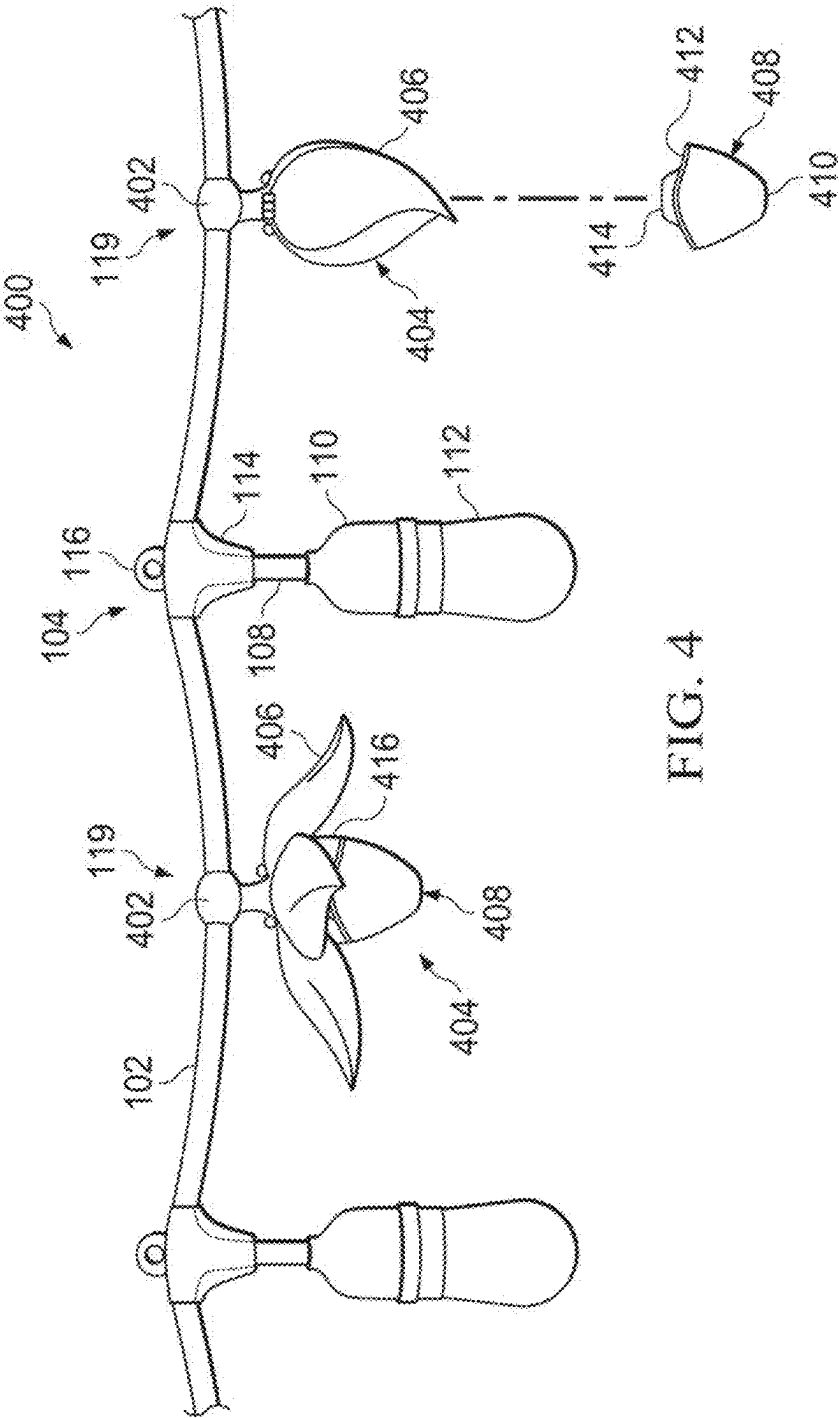
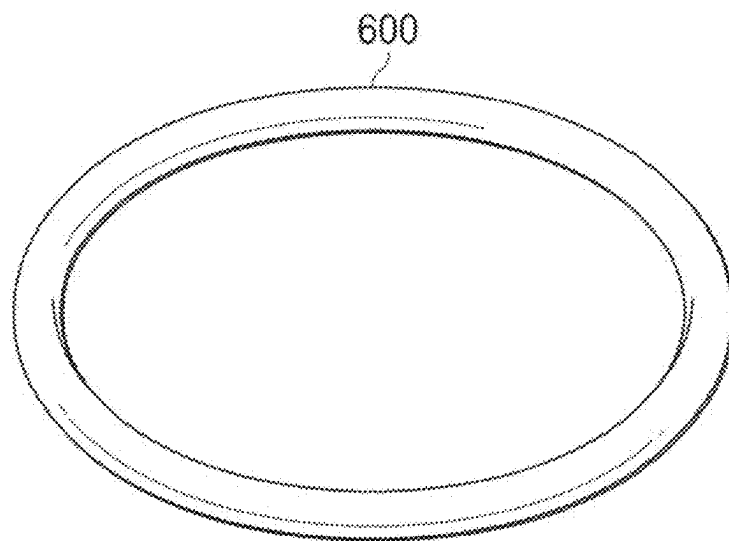
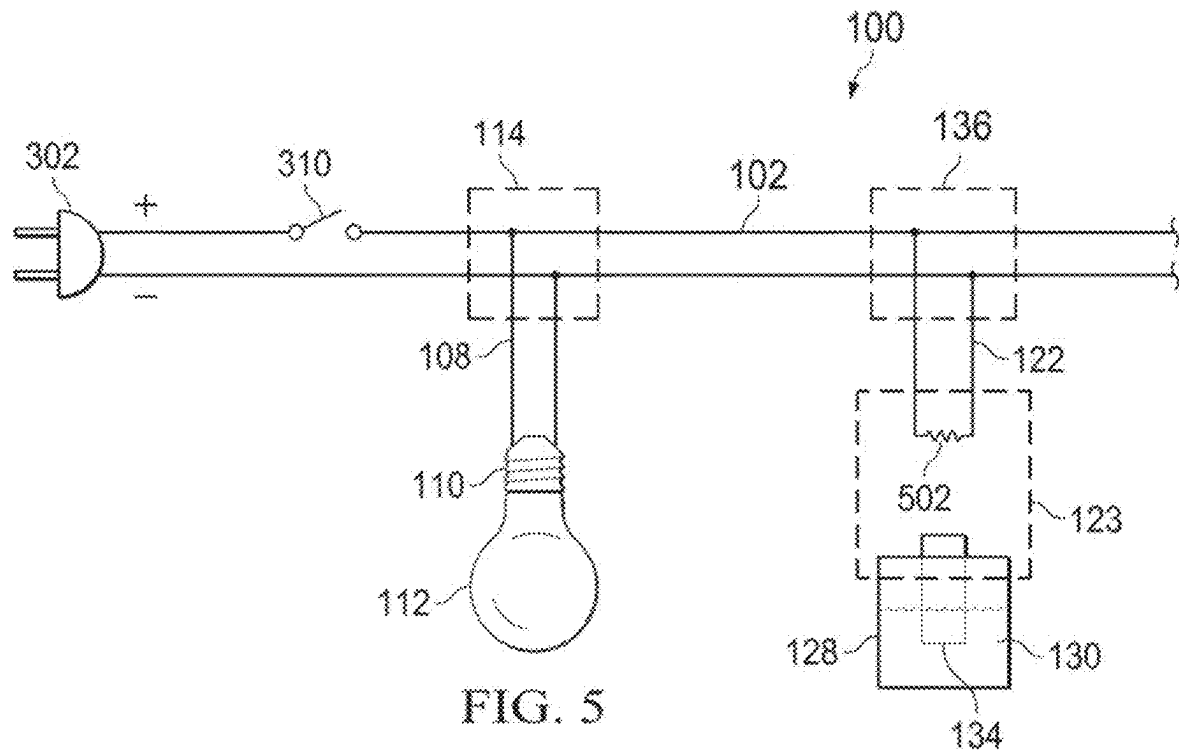
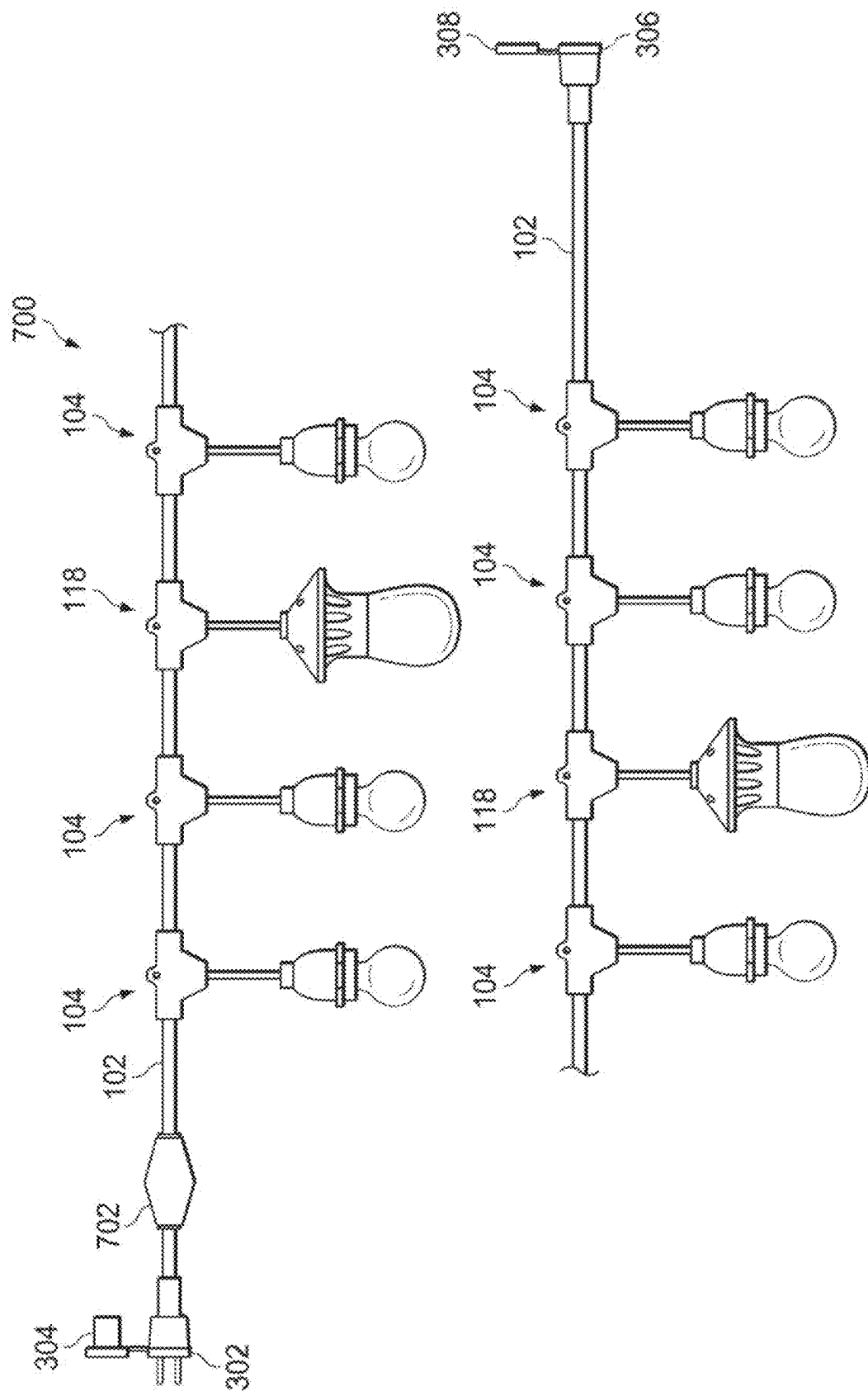


FIG. 2









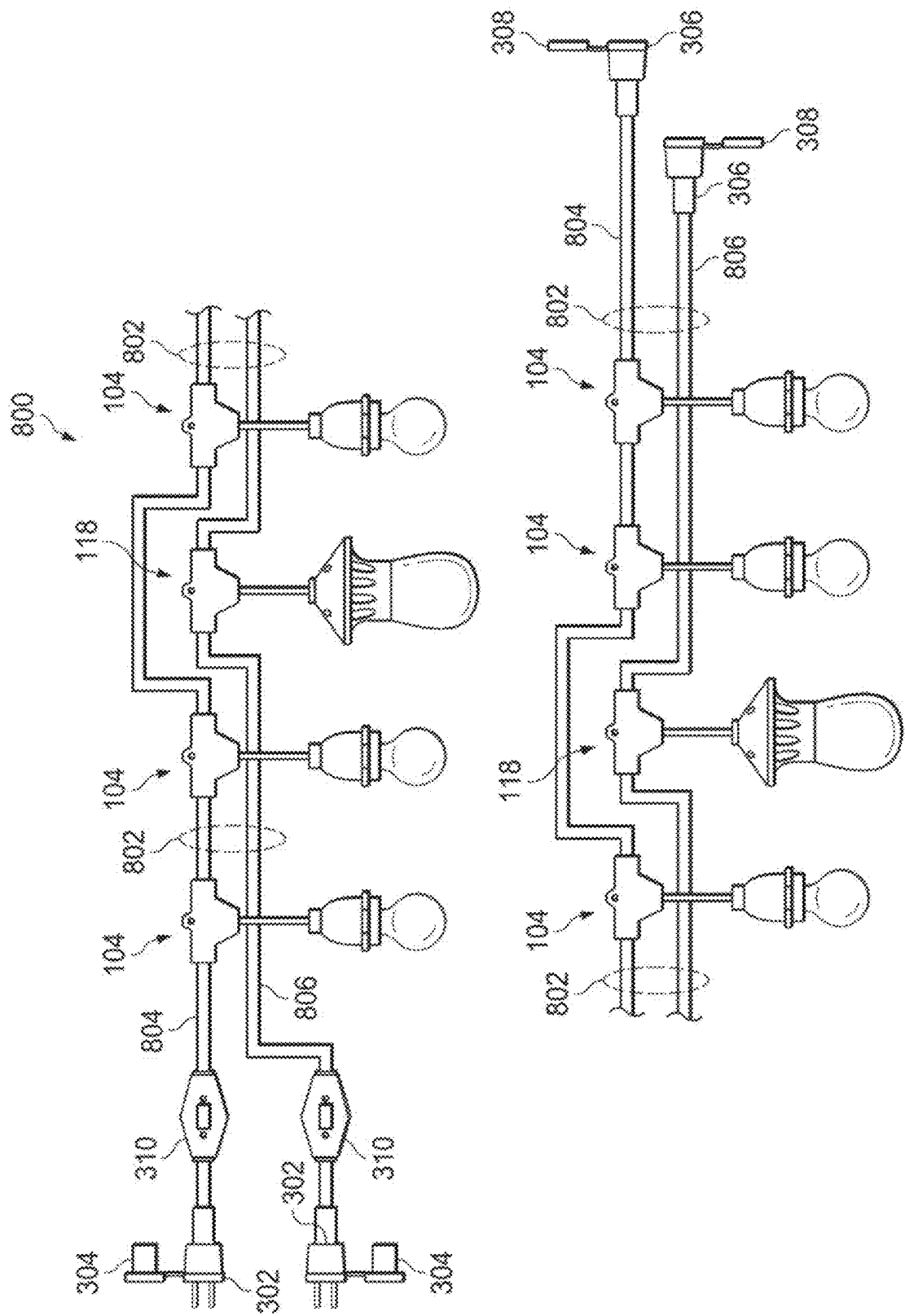


FIG. 8

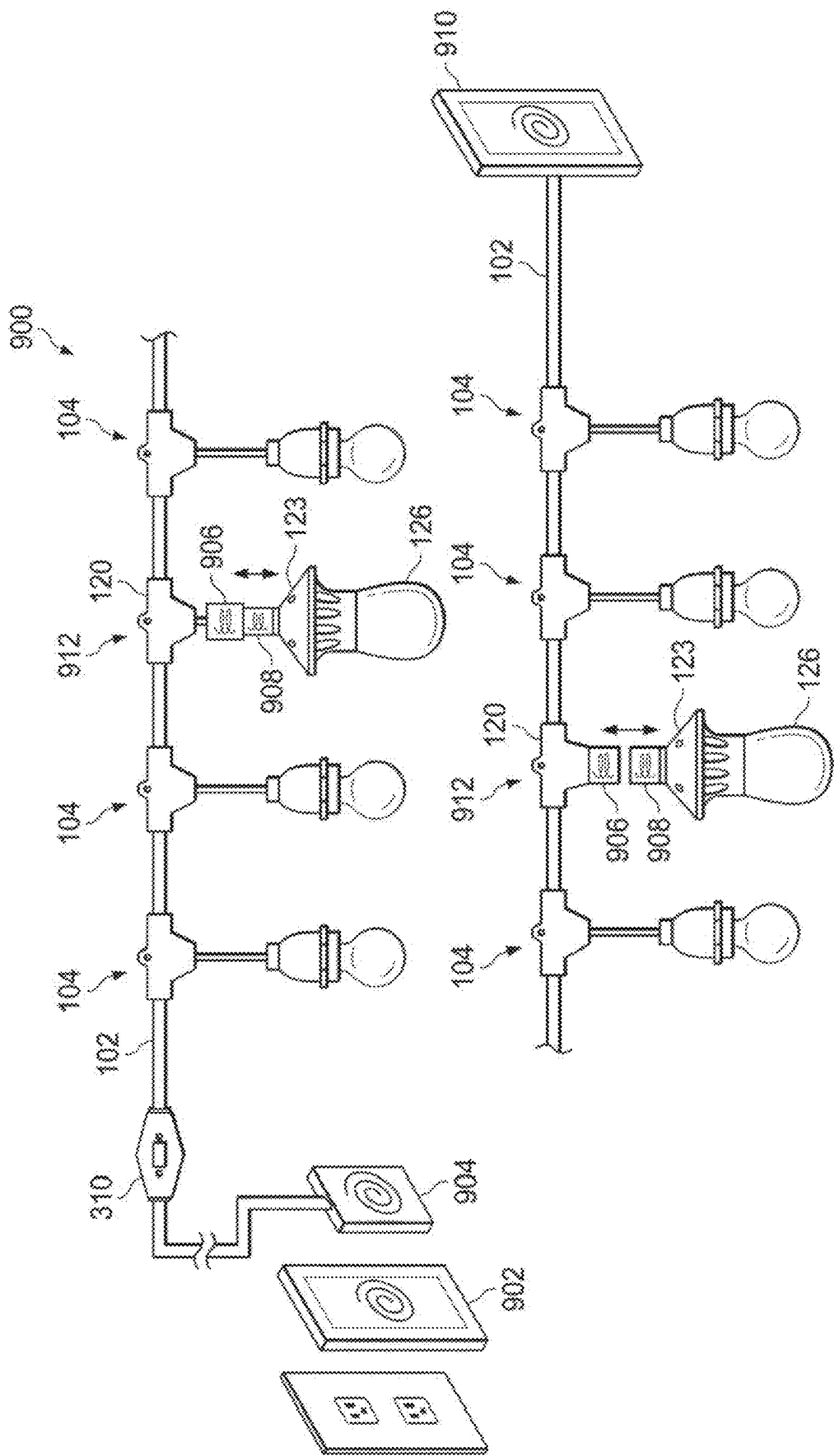
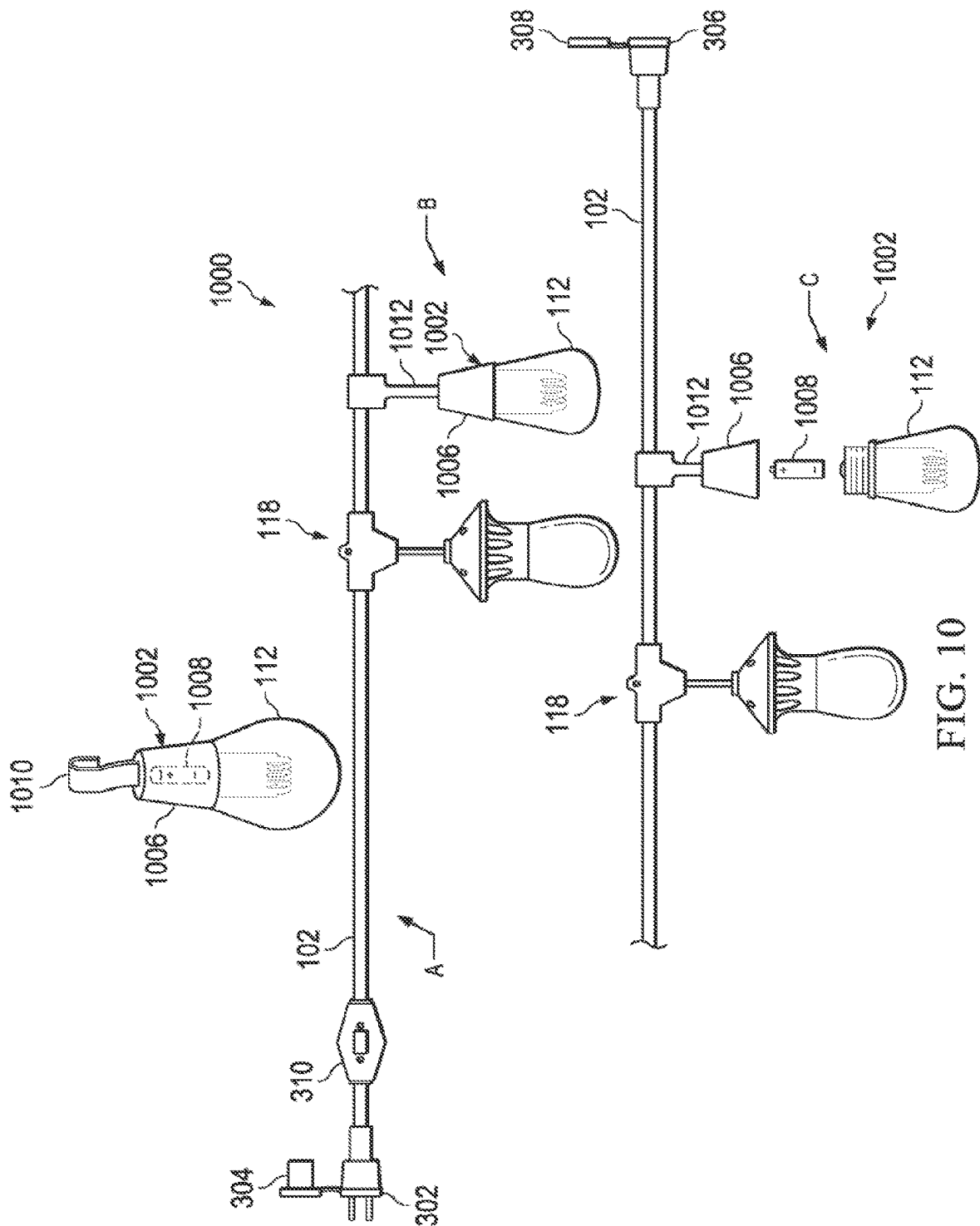
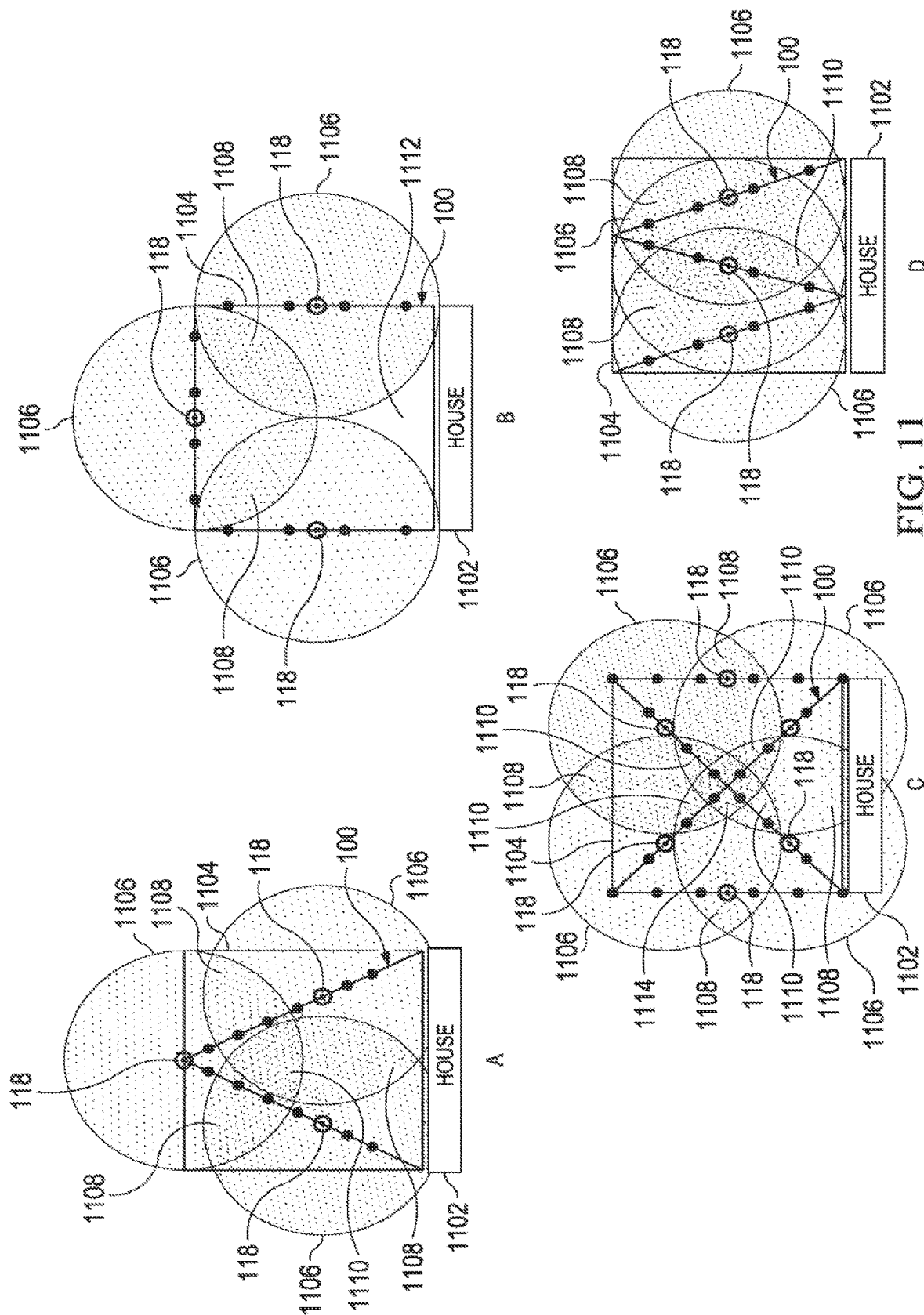


FIG. 9





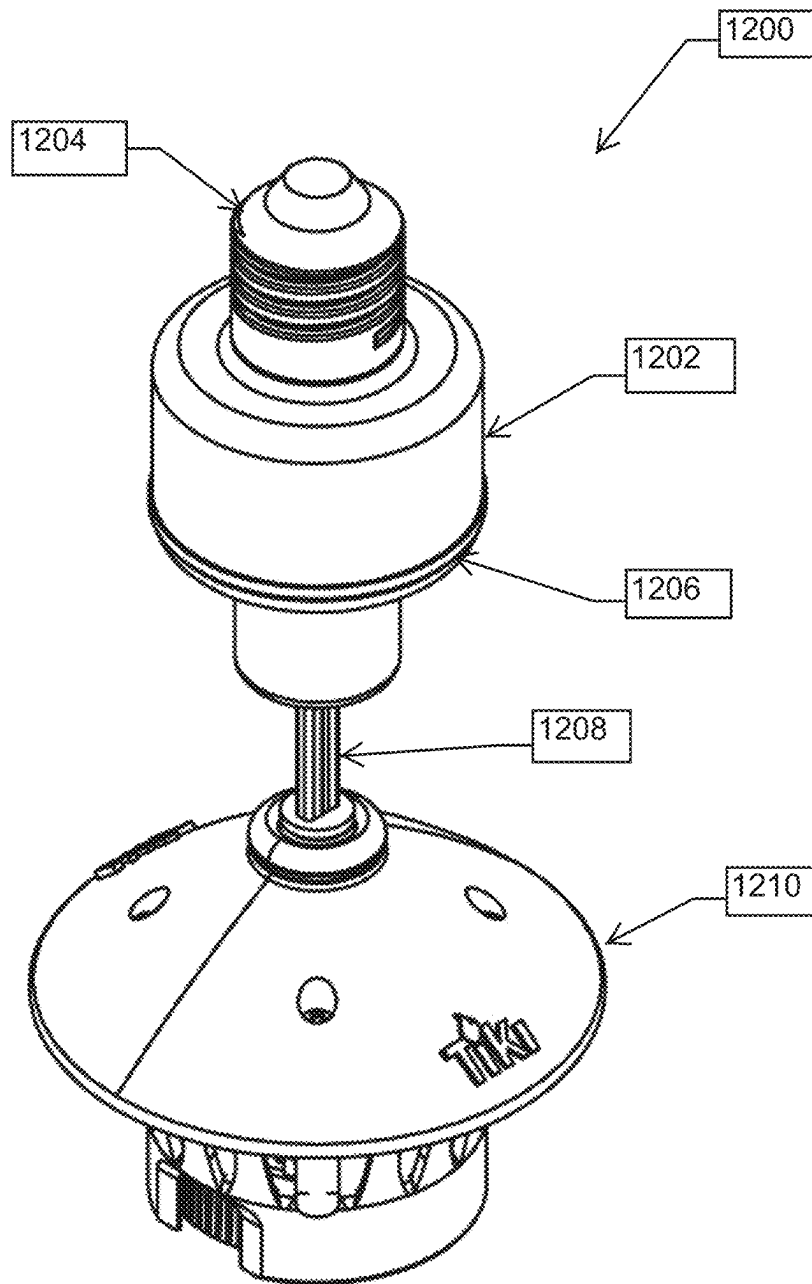


Figure 12A

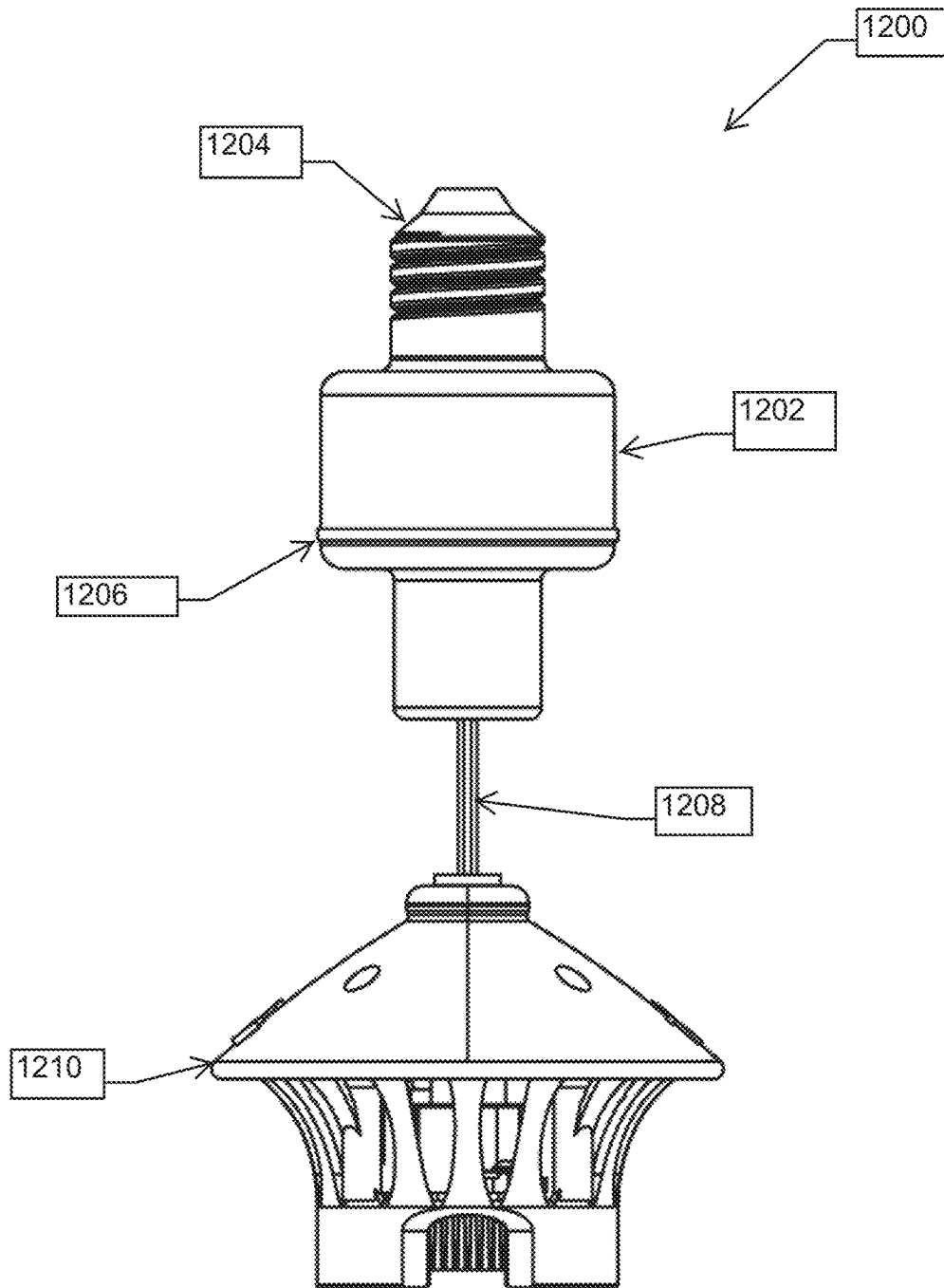


Figure 12B

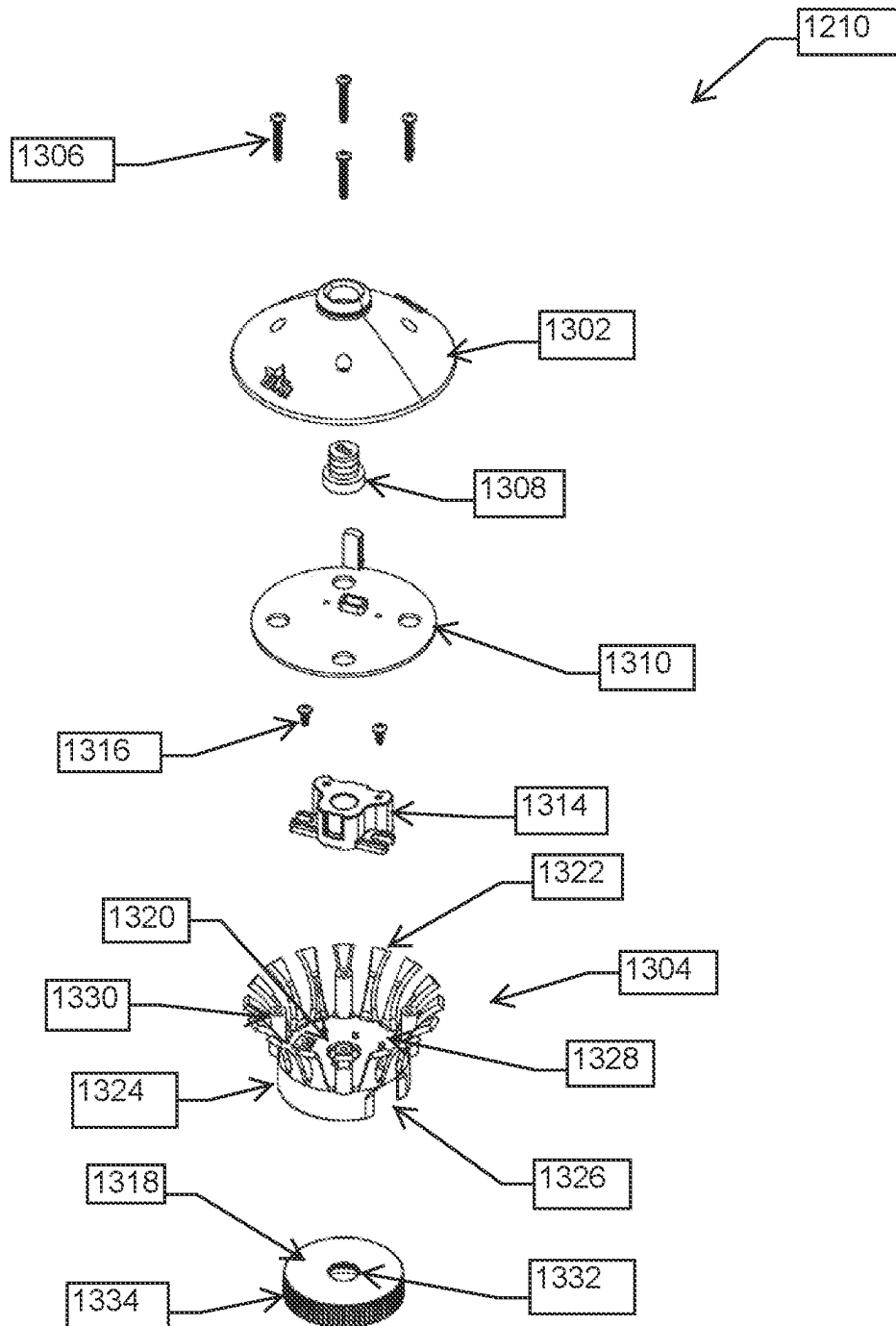


Figure 13A

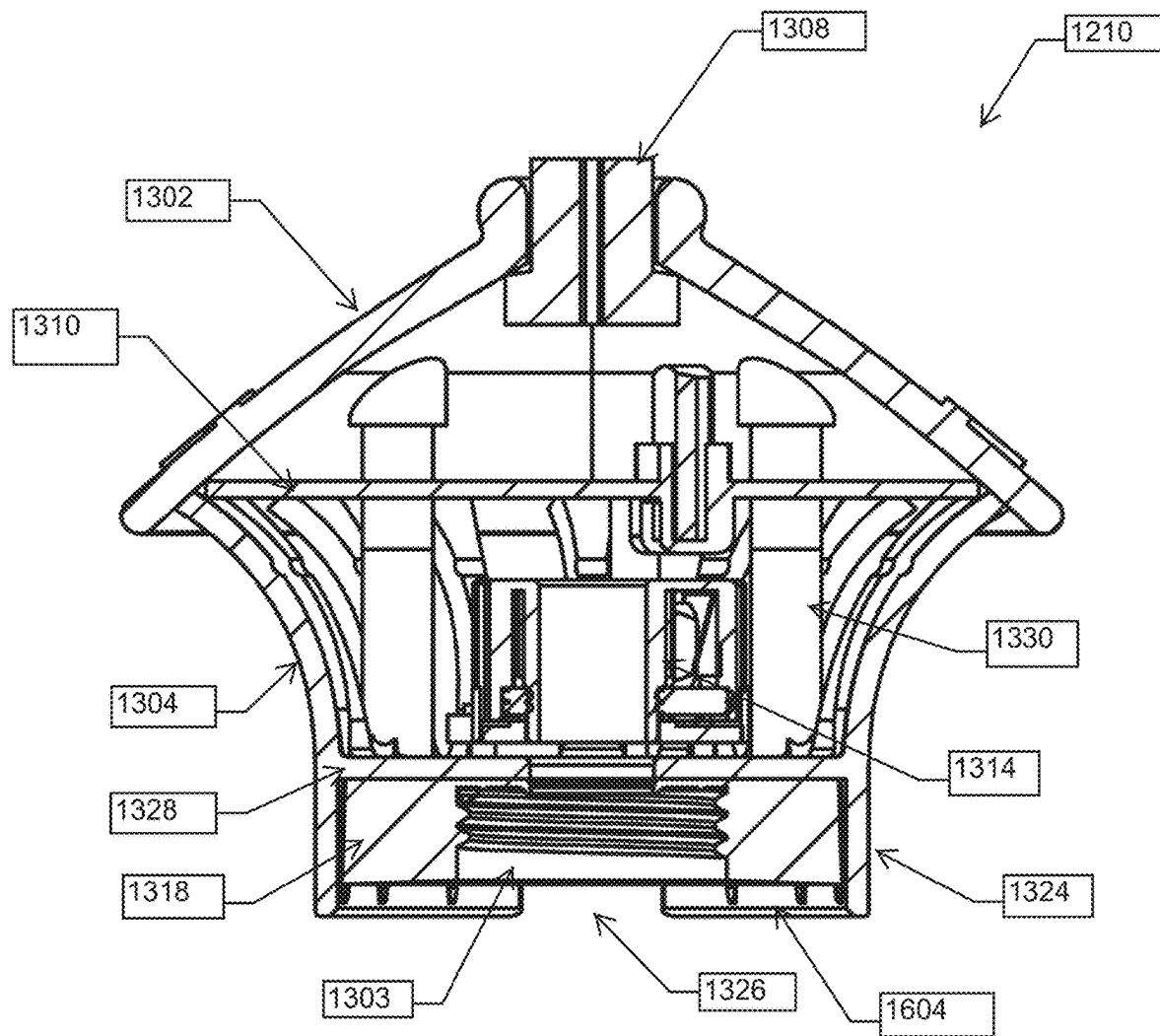


Figure 13B

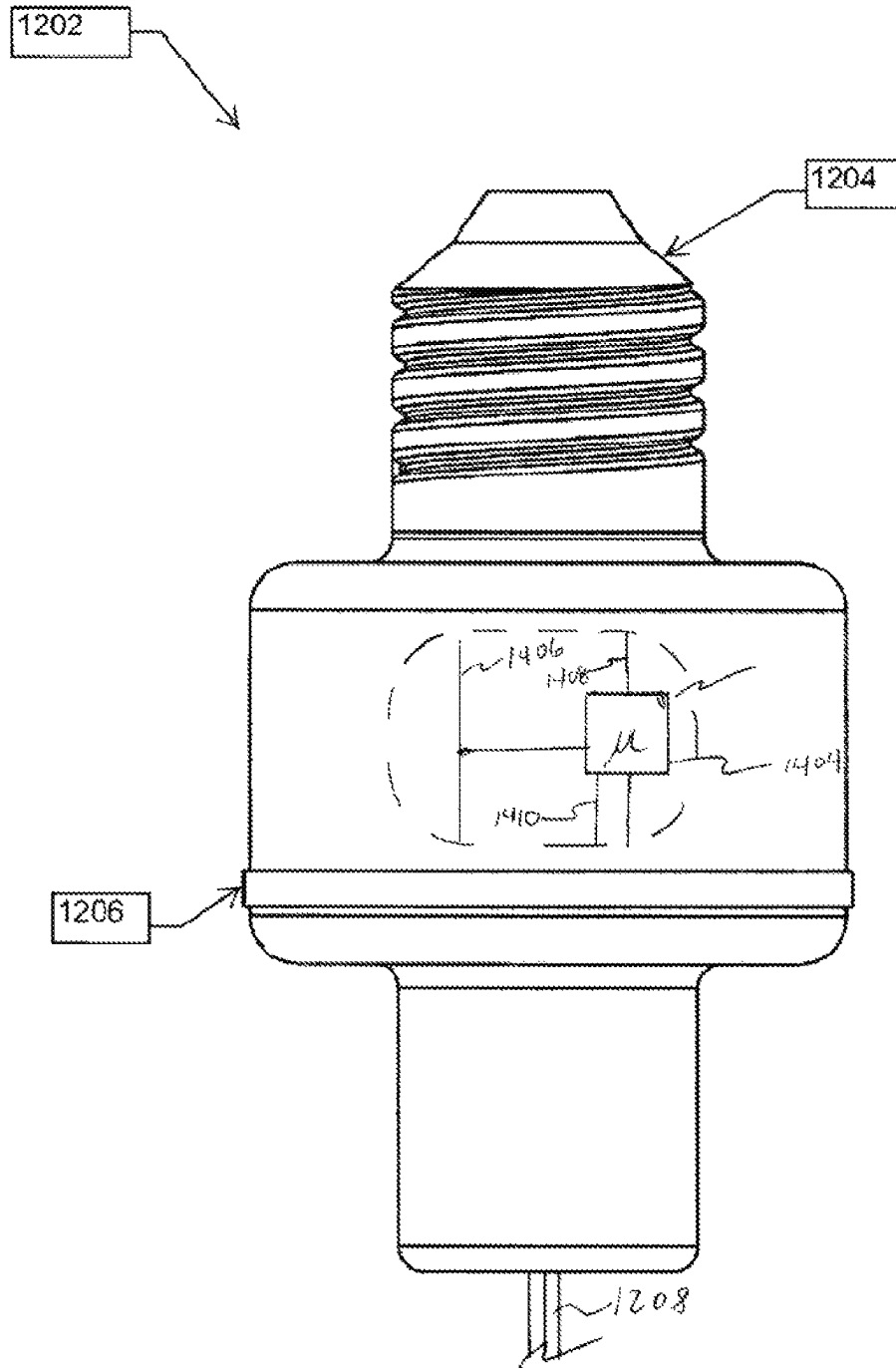


Figure 14

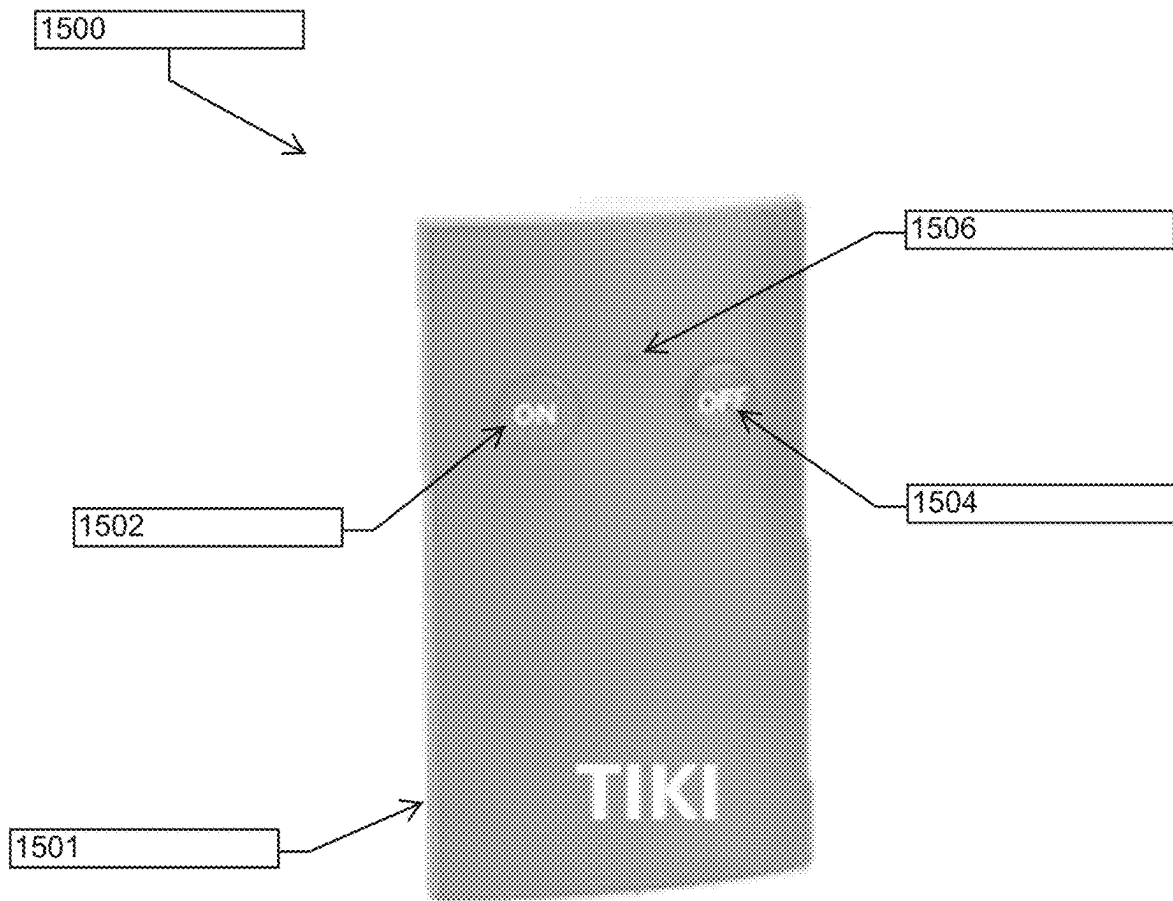


Figure 15

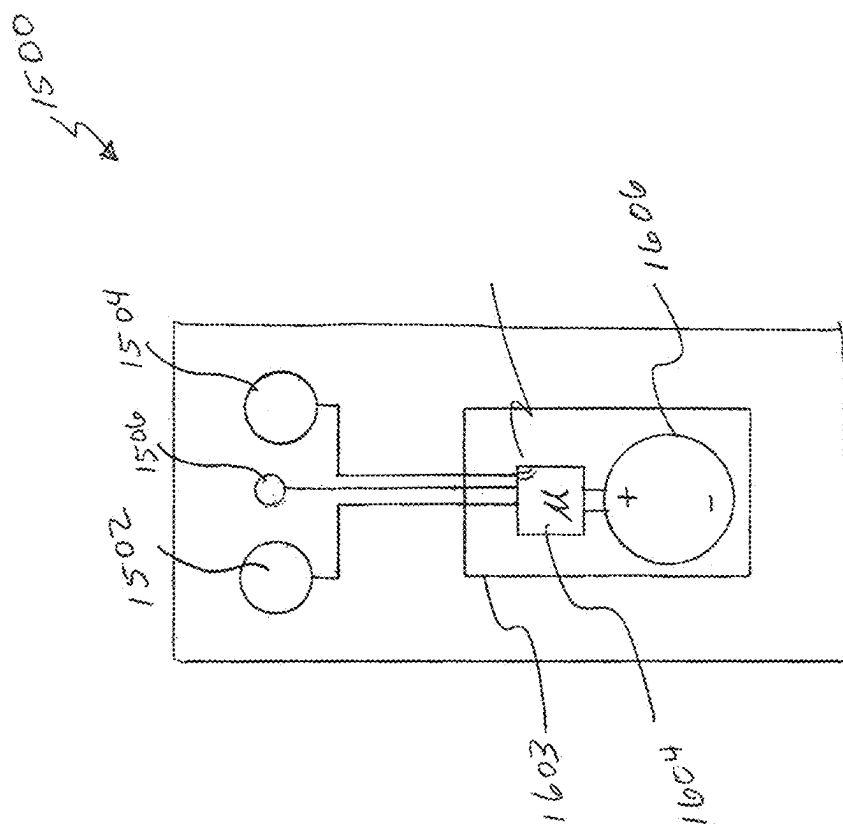


Fig. 16

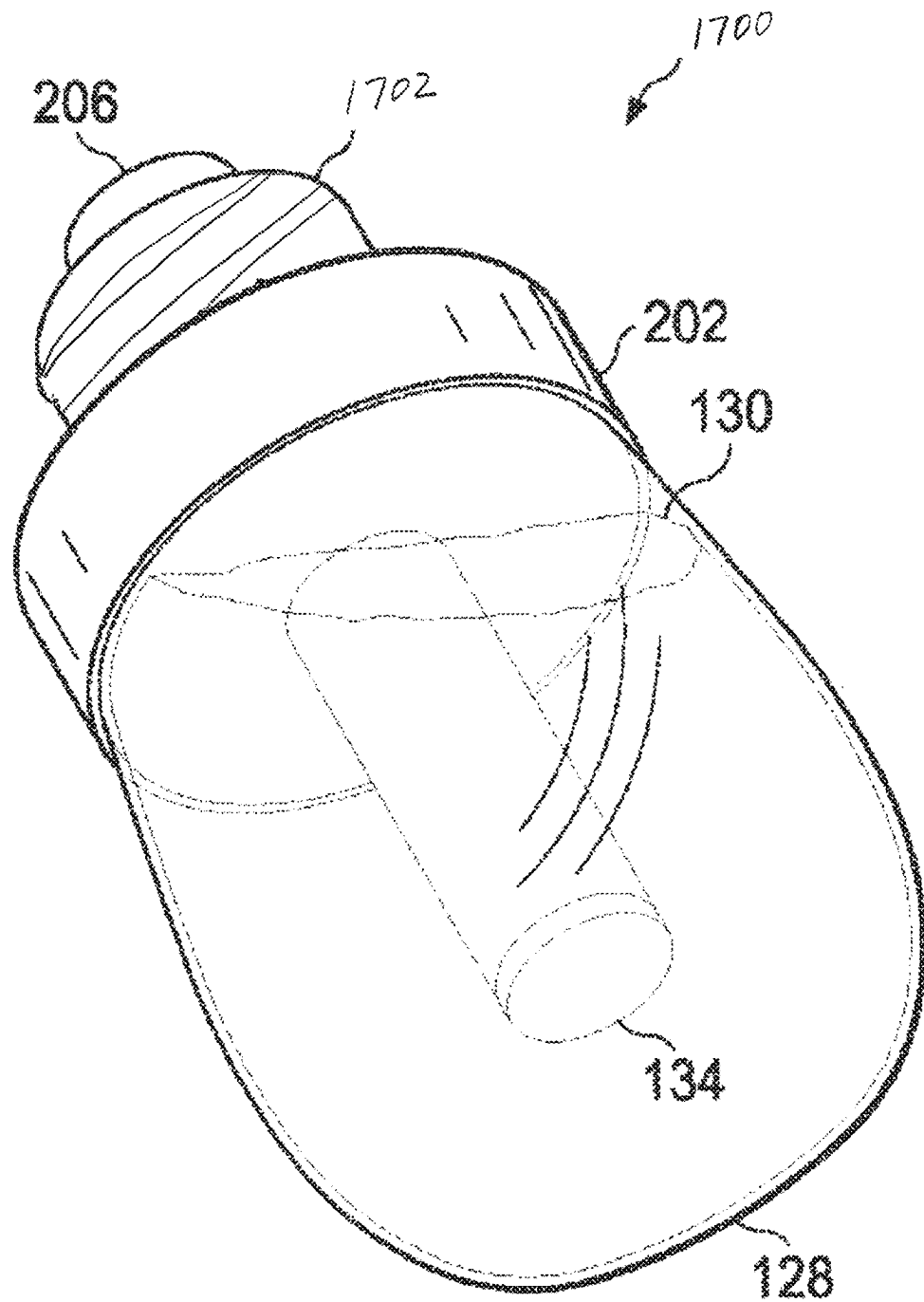


Figure 17

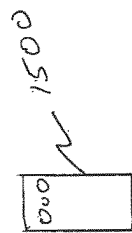
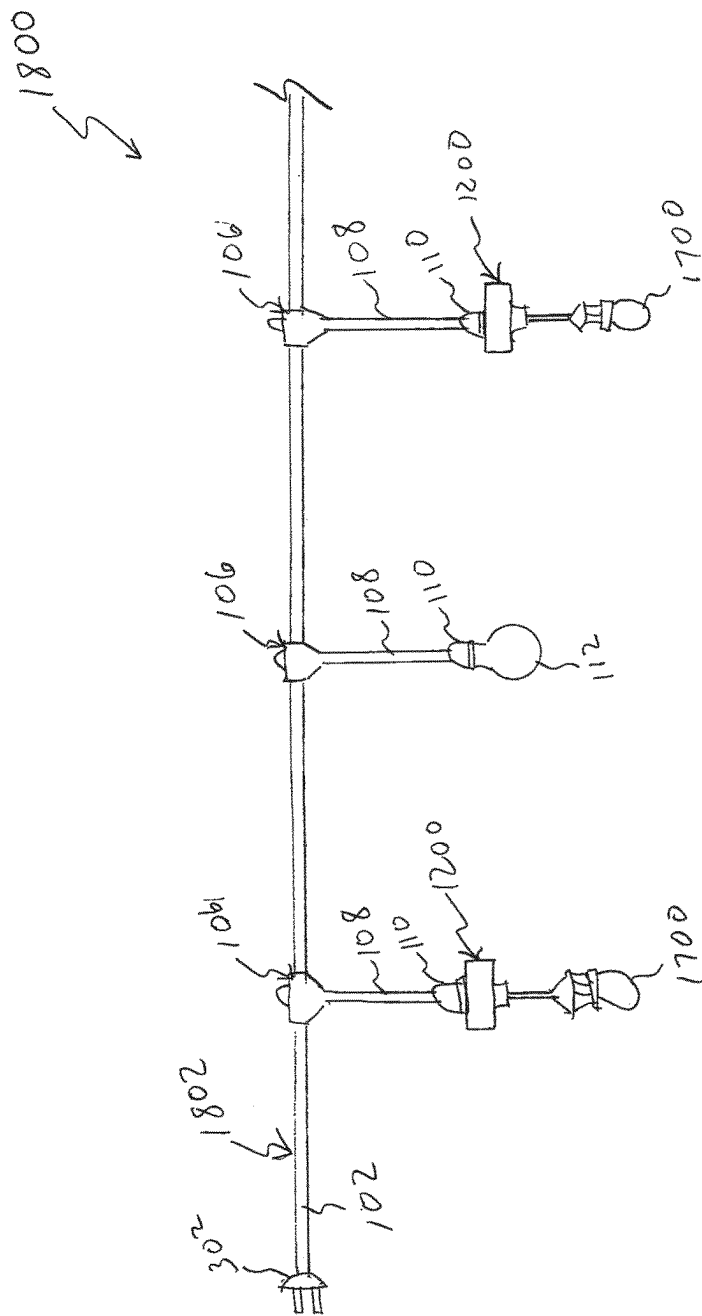


Fig. 18

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REPELLANT STRING LIGHT AND RETROFIT ASSEMBLY

CROSS-REFERENCE TO RELATED CASES

This application is a continuation in part which is continuation of U.S. patent application Ser. No. 17/101,827 filed on Nov. 23, 2020, which is a continuation-in-part of U.S. Pat. No. 10,962,219 filed on Aug. 5, 2019, which claims the benefit of U.S. provisional patent application Ser. No. 62/714,347, filed on Aug. 3, 2018 entitled REPELLANT STRING LIGHT, the contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

This disclosure relates to outdoor devices having multiple uses. In particular, but not by way of limitation, this disclosure relates to devices providing light and dispersal of scents and/or repellants.

BACKGROUND OF THE INVENTION

Outdoor party lights or string lights have become common for lighting or decorative purposes. These may be based upon low wattage lights such as LEDs (light emitting diodes). However, their utility has remained static for some time and they have not generally been put to further uses.

Dispersion of repellants via electrical power, or plug-in devices, can enhance delivery of effective repellants. However, utility may be limited where battery power is required or where location of use has been restricted to locations very near an outlet. Additionally, for maximum effect with a repellant, it should be placed where it has the greatest exposure to the area sought to be protected.

What is needed is a system and method for addressing the above and related concerns.

SUMMARY OF THE INVENTION

The invention of the present disclosure, in one aspect thereof, comprises a repellant assembly including a receiver housing with a light socket connector, a heating receptacle electrically connected to the receiver housing, the heating receptacle having an internal electric heater and a fitting for selectively receiving a repellant pod assembly in a position proximate the electric heater to heat the repellant pod assembly to disperse repellant, and a remote-controlled switch in the receiver housing that switches power from the light socket connector to the heating receptacle.

In some embodiments, the heating receptacle further comprises a body defining a sleeve, and a collar rotatably fitted into the sleeve and defining the fitting for selectively receiving the repellant pod assembly in the position proximate the electric heater to heat the repellant pod assembly to disperse repellant. A window may be defined in the sleeve allowing user contact with the collar. The collar may have a textured outer circumferential surface contactable through the window.

In some cases, a length of power cord connects the heating receptacle to the receiver housing.

The remote-controlled switch may comprise a solid-state switch. The solid-state switch may comprise a wireless communications capable microcontroller. The repellant assembly may include a remote control to provide communication to the microcontroller to power on and power off the heating receptacle.

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The invention of the present disclosure, in another aspect thereof, comprises a repellant assembly having a light socket connector, a heating receptacle that selectively connects to a replaceable repellant pod that is selectively heated to disperse repellant from the repellant pod, and a remote controlled switch interposing the light socket connector and the heating receptacle to power the heating receptacle on and off.

The heating receptacle may further comprise a body having a top connected to a base to define an interior, a body floor with a sleeve descending therefrom, the sleeve providing a fitting for connecting to the replaceable repellant pod, an aperture defined in the body floor, and an electric heater affixed near the aperture in the body floor.

One of the body top and body base may be ventilated to allow repellant to escape from the interior. The sleeve may contain a collar rotatably fitted therein, the collar defining a threaded connection proximate the aperture in the body floor. The sleeve may define an opening that allows user contact with the collar and the collar defines an outer textured circumferential surface proximate the opening. The collar may retain the replaceable repellant pod in the threaded connection such that a wick from the replaceable repellant pod is held proximate the electric heater.

In some embodiments, the repellant assembly further comprises a receiver housing that contains the remote-controlled switch. The remote-controlled switch may comprise a solid-state device that receives wireless commands from a wireless remote control. The repellant assembly may further comprise an indicator on the receiver housing that is illuminated by the remote controlled switch when the heating receptacle is powered on.

The invention of the present disclosure, in another aspect thereof, comprises a repellant system including a power cord having a plug for an outlet, a plurality of light sockets powered by and attached to the cord, and at least one repellant dispersion port supplying power provided by the power cord to a resistive heating element in proximity to a repellant contained in a fluid pod, the repellant delivered from the bulb for evaporation by the resistive heating element by a wick placed into the repellant in the fluid pod. A dispersal of the repellant provides a repellant zone from the at least one repellant dispersion port to a ground level below repellant dispersion port.

In some embodiments, the at least one repellant dispersion port connects to the power cord via one of the plurality of light sockets. The at least one repellant dispersion port may be activated and deactivated by remote control.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a close-up elevational view of a portion of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 2 is a perspective view of a repellant assembly according to aspects of the present disclosure.

FIG. 3 is an elevational view of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 4 is a close-up perspective view of another embodiment of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 5 is a simplified electrical schematic diagram of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 6 is a perspective view of a repellant level indicator float ring according to aspects of the present disclosure.

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FIG. 7 is an elevational view of another embodiment of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 8 is an exploded plan diagram of another embodiment of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 9 is an elevational view of another embodiment of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 10 is an exploded plan diagram of another embodiment of a lighting and dispersal system according to aspects of the present disclosure.

FIG. 11A is an overhead view of a plurality of repellent zones arranged in first plan according to aspects of the present disclosure.

FIG. 11B is an overhead view of a plurality of repellent zones arranged in first plan according to aspects of the present disclosure.

FIG. 11C is an overhead view of a plurality of repellent zones arranged in first plan according to aspects of the present disclosure.

FIG. 11D is an overhead view of a plurality of repellent zones arranged in first plan according to aspects of the present disclosure.

FIG. 12A is a perspective view of an embodiment of a retrofit assembly according to aspects of the present disclosure.

FIG. 12B is a side view of the retrofit assembly of FIG. 12A.

FIG. 13A is an exploded view of a heating receptacle according to aspects of the present disclosure.

FIG. 13B is a cutaway view of a heating receptacle of FIG. 12A.

FIG. 14 is a side, partial cutaway view of one embodiment of a socket connector for use with systems of the present disclosure.

FIG. 15 is perspective view of a one an embodiment of a wireless remote control for use with systems of the present disclosure.

FIG. 16 is a simplified cutaway view of the wireless remote control of FIG. 15.

FIG. 17 is a perspective view of a repellent assembly pod for use with systems of the present disclosure.

FIG. 18 is a side view of a retrofitted light string according to aspects of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Referring now to FIG. 1, a close-up elevational view of a portion of a lighting and dispersal system 100 according to aspects of the present disclosure is shown. The system 100 is a combination light string and liquid volatilization and dispersal mechanism. Systems of the present disclosure may disperse repellants, scents, oils, chemicals etc., while providing lighting or decorative effects. The system 100 provides a power cord 102 with a plurality of light receptacles 104 spaced along the cord 102. The light receptacles 104 may further each comprise a cord junction 106 with a segment of transverse power cord 108 leading from the junction to a light socket 110. In some embodiments, the light socket 110 is affixed directly to the cord junction 106 (i.e., there is no transverse power cord 108).

The junction 106 provides an internal electrical splice to the power cord 102 which may power a light bulb 112 inserted into the socket 110. The junction 106 may comprise a body 114 that may be constructed of an insulating polymer

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or another material. The junction 106 may comprise a hang loop 116 defined in or attached to the body 114.

In various embodiments, the light socket 110 may comprise a screw in base such as an E26/E27 base or another base such as E12, E14, E17, E39/E40. In some embodiments, the light socket 110 may comprise a bayonet or press-in style socket. The light bulb 112 may comprise an incandescent light bulb, a compact fluorescent (CFL) bulb, or a light emitting diode (LED) in a form factor to fit the associated socket 110. In one embodiment, the light bulb 112 comprises a 2-watt LED.

In addition to one or a plurality of light receptacles 104, the system 100 comprises one or a plurality of repellent dispersion ports 118. The repellent dispersion ports 118 may each comprise a cord junction 120 connecting a transverse power cord segment 122 to a heating receptacle 123. The cord junction 120 comprises a body 136 covering an electrical splice from the power cord 102 to supply power to the heating receptacle 123. The body 136 may comprise an insulating polymer. In some embodiments, the heating receptacle 123 is affixed directly to the cord junction 120 (i.e., there is no transverse power cord 122).

The heating receptacle 123 may attach selectively to a repellent assembly pod 126. The pod 126 may removably inserted into or attach to the receptacle 123 with a threaded connection, a turn and lock connected, a press fit mechanism, a bayonet style connected, a magnetic fitting, or another operable mechanism. The heating receptacle 123 applies heat to the repellent assembly pod 126 to volatilize and disperse the contents of the pod 126. To that end, ventilation ports 125 may be defined in a body 124 comprising the heating receptacle 123. The body 124 may comprise a polymer or other suitable material.

The pod 126 may comprise a fluid reservoir 128 that contains a quantity of a pest or insect repellent product 130 such as a synthetic pyrethroid, metofluthrin, meperfluthin, or other product. The fluid reservoir 128 may also contain scents or other agents. The pod 126 may be provided to the consumer pre-filled and may or may not be serviceable for refilling by the consumer (e.g., the pod 126 may be considered a disposable item). The fluid reservoir 128 may comprise an opaque or tinted polymer. In some embodiments, the fluid reservoir 128 may be clear or at least translucent to allow the consumer to ascertain the amount of repellent 130 remaining in the reservoir 128.

In some cases, various lighting conditions and combinations of the appearance of the repellent 130 and the fluid reservoir 128 may render determination of the remaining amount of repellent 130 difficult. In some embodiments, a number of floats 132 may be provided in the repellent 130. In the case of small discrete floats 132 or beads, surface tensions of the repellent 130 and the geometry of the meniscus that forms at the top of the repellent layer results in the floats 132 tending to remain near the sides of the repellent bulb 128 such that the fluid level may be easily determined. In other embodiments, a toroidal ring-type float (600, FIG. 6) may be utilized.

Referring now also to FIG. 2 a perspective view of the repellent assembly 126 removed from the heating receptacle 123 is shown. It can be seen that no floats or fluid level indicators are provided in the view of FIG. 2 and that the fluid reservoir 128 is substantially full as it would appear before use (or after refill). A cap 202 may affix to the top of the fluid reservoir 128 to prevent spillage of the repellent 130. The cap 202 may provide a fitting 204 for selective attachment into the heating receptacle 123. Here the fitting 204 is illustrated as a threaded fitting. A top port 206 may be

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provided for passage or evaporation of the repellant **130**. In one embodiment, a top of the wick **134** serves as a top port or evaporation pad.

Referring now to FIG. **3** an elevational view of the lighting and dispersal system **100** is shown. FIG. **3** is not to scale, nor does it necessarily reflect the total number of light receptacles **104** and repellant dispersion ports **118** associated with the system **100**. In some embodiments, there may be equal numbers of each. However, the repellant dispersion ports **118** may provide a wide enough effective repellant area that a relatively greater number of light receptacles **104** may be appropriate.

In one embodiment, power cord **102** may be around 36 feet in length. A total of three repellant dispersion ports **118** are provided along the power cord **102** with a total of twelve light receptacles **104**. Various embodiments may be provided with a standard outlet plug **302** (e.g., a standard US/Canada or other type of plug), possibly with an attached cover **304**. At an opposite end a standard female outlet **306** may be provided, possibly with an attached cover **308**. A power switch **310** may be provided as well. The outlet **306** allows for multiple systems **100** to be chained together, or for the system **100** to provide “downstream” power to other devices.

Referring now to FIG. **4**, a close-up perspective view of another embodiment of a lighting and dispersal system **400** according to aspects of the present disclosure is shown. The system **400** shares many components in common with the system **100** previously described. A power cord **102** provides a common source of power. One or more light receptacles **104** may be provided at various locations along the power cord **102**. The power cord may have electrical connections at either end (not shown) as well as a power switch (not shown).

The system **400** comprises one or more repellant dispersion ports **119** that differ in some respects from the repellant dispersion ports **118** discussed previously. The repellant dispersion ports **119** comprise junctions **402** providing an internal power splice and affixing the repellant dispersion port **119** to the power cord **102**. Here, the junction **402** also provides a heating element for promotion of evaporation of the associated repellant and other chemicals. A covering **404** selectively surrounds a repellant assembly **408** and heating receptacle **416**. In some embodiments, the covering **404** resembles a flower with openable petals **406**. The petals **406** may be opened manually to reveal the repellant assembly **408**, they may open under a power mechanism, or they may be configured with a low enough mass and loose enough connections to be able to displace under light breezes to reveal the repellant assembly **408**.

The repellant assembly **408** may contain any of the aforedescribed repellants or other ingredients, and may be refillable or disposable. As show in the inset, the repellant assembly **408** may include a fluid pod **410** that contains the repellant or other ingredients. The fluid pod **410** may be provided with a cap **412** that may provide a fitting **414** for attachment and detachment from the heating receptacle **416**. A wick (not visible) may provide a pathway for movement of the repellant from the fluid pod **410** for dispersion/evaporation by the heating receptacle **416**.

Referring now to FIG. **5** a simplified electrical schematic diagram of a lighting and dispersal system according to aspects of the present disclosure is shown. FIG. **5** corresponds particularly to the system **100**. However, those of skill in the art will appreciate that the schematic is easily adapted to the system **200**, or any of the additional embodiments discussed herein. The plug in **302** is shown providing

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power to the power cord **102** shown as positive and negative leads. Switch **310** controls the positive side of power cord **102**. Junction body **114** contains the connecting nodes for transverse power cord **108** to supply current and voltage to light socket **112** and bulb **112**.

Junction body **136** contains nodes connecting power cord **102** to transverse power cord **122** leading to the heating receptacle **123**. A resistive heating element **502** exposes wick **134** (or at least the repellant) to elevated temperature to gasify or evaporate the same. In some embodiments the wattage of the resistive heating element **502** is less than 8 watts. In another embodiment, the wattage of resistive element **502** is 5 watts or about 5 watts. The temperature provided by the resistive heating element **502** may be from about 120 C to about 140 C.

It should be understood that the arrangement of the components shown in FIG. **5** may be replicated as far as necessary to properly configure each associate lighting receptacle and repellant dispersion port. The system **100** may be configured with the appropriately rated components to operate from about 100V to about 240V. It should also be understood that, in some embodiments, lighting and repellant functions are isolated along the power cord **102**. In other words, in some embodiments, the light receptacles **104** are divorced from dispersion of any repellants or other chemicals and are, instead, dedicated to providing only light. Correspondingly, in some embodiments, repellant dispersion ports **118** do not provide any useable light.

In operation, the systems described herein (e.g., system **100** and system **400**) may be installed such that they are somewhere above ground level to allow maximum effectiveness of the repellant. In various embodiments 4 feet to 12 feet provides an optimum effect and a repellant zone from mounting level all the way to the ground. The systems may be strung from a roof, fence, wall, tree, dedicated poles, or other structures. The systems **100**, **400** are also described as being powered by a household outlet. However, it should be understood they may also be powered by batteries, generators, and other portable power sources.

Referring now to FIG. **7**, an elevational view of another embodiment of a lighting and dispersal system **700** according to aspects of the present disclosure is shown. The system **700** is similar to the system **100** described above but does not have a manual power switch (such as power switch **310**). The system **700** may power on in response to being connected to power, via plug **302**, for example. In some embodiments, a delay circuit, warmup circuit, current limiting, circuit and/or power conditioning circuit as known in the art may be provided in a circuit enclosure **702**. Some embodiments to not provide such circuitry nor the enclosure **702** but such circuitry may still be included with the specific light receptacles **104** and/or repellant dispersion ports **118** if and as needed. Fuses, as known in the art, may also be include if and where needed.

Referring now to FIG. **8** an exploded plan diagram of another embodiment of a lighting and dispersal system **800** according to aspects of the present disclosure is shown. The system **800** comprises both lighting and repellent dispersion on a single cord **802** but the functions of both are electrically separated or isolated. The system **800** comprises a first power line **804** that supplies power for a plurality of light receptacles **104** and a second power line **806** that supplies power for a plurality of repellant dispersion ports **118**. The lines **804**, **806** may be physically bound and/or wound together along medial portions thereof. They may have a separate outer wrapper (not shown) for weather resistance, insulation, appearance, and/or ease of handling. At ends of

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the lines **804**, **806** they may split apart into separate plugs **302** and may separate into a pair of receptacles or outlets **306**. Cord junctions (e.g., **106**, **120**, FIG. 1) and/or hang loops (e.g., **116**, FIG. 1) and possibly other components may be available outside any wrapper or covering of the power lines **804**, **806** as needed.

In some embodiments, each line **804**, **806** has its own manual switch **310** such that dispersion and lighting functions are separately controllable. In other embodiments, one or both of the power lines **804**, **806** may be provided with no manual switch **310** such that they are controlled by connection to power. In such case, either line **804**, **806** may be provided with whatever adjunct circuitry may otherwise be needed, as with the system **700** (e.g., delay circuit, warmup circuit, current limiting, circuit and/or power conditioning, etc.).

FIG. 9 is an elevational view of another embodiment of a lighting and dispersal system **900** according to aspects of the present disclosure. The system **900** is similar to the system **100** discussed above but utilizes wireless power couplings. It should also be understood that wireless power couplings can be utilized with any system according to the present disclosure. As illustrated, the system **900** may utilize a plug-in coil **902** that connects to household power. A line coil **904** is placed onto, in, or near the line coil **904** such that electric power is transferred thereto, and into the power cord **102**. It should be understood that any wireless power transfer technology as is known in the art may be employed in this context. Nonlimiting examples include inductive, capacitive, and magnetodynamic coupling.

Dispersion ports **912** may also be adapted to utilize wireless power transfer but may otherwise operate similarly to dispersion ports **118** described above. For example, the ports **118** as shown in FIG. 8 utilize an inner coil coupling **908** affixed to heating receptacle **123** that fits into an outer coil coupling **906** that is electrically connected to the power cord **102** (possibly via cord junction **120**). In the illustrated embodiment, the couplings **906**, **908** provide for wireless power transfer from the power cord **102** to the receptacle **123** and also physically retain the receptacle **123** and associated pod **126** in place on the cord **102** and/or junction **120**. Wireless power transfer may be any suitable known protocol or connection type (e.g., magnetic, inductive, etc.). The physical retention may be via interference fit, friction fit, securement latch, bayonet coupling, screw coupling, other mechanism.

Opposite from the line coil **904** on the power cord **102** may be an outlet plug in **910**. This may extend the output provided by plug in coil **902** such that additional light/repellant strings or other devices can be daisy chained to the system **900**. The system **900** may or may not include a manual power switch **310** and/or other power conditioning and needed operational circuitry as is known in the art.

It should also be understood that the light receptacles **104** may be configured to utilize wireless power couplings similar to the manner in which dispersion ports **118** (e.g., FIG. 1) are configured as dispersion ports **912**.

Referring now to FIG. 10, an exploded plan diagram of another embodiment of a lighting and dispersal system **1000** according to aspects of the present disclosure is shown. The system **1000** utilizes a power cord **102** (which may be equipped as in FIG. 3, or according to any other embodiment described herein) that has only repellent dispersion ports **118** rather than dispersion ports **118** and light receptacles. In another embodiment, the system **1000** has a cord **102** provided with light receptacles as well (e.g., **104**, FIG. 1) but additional lighting is desired. The additional lighting may be

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the same or different from that provided via the receptacles **104** described above and may be provided via one or more clip-on lights **1002**.

The clip-on lights **1002** may comprise a light bulb **112** as described above. The bulb **112** may be selectively retained and provided power by a housing **1006** with an internal battery **1008**. The battery **1008** may be any battery providing a suitable voltage for the bulb **112** (whether LED or otherwise) and may be based on any suitable chemistry (e.g., alkaline, lithium ion, etc.). In some embodiments, the battery **1008** is rechargeable.

The housing **1006** has a clip **1010** affixed thereto that selectively retains the clip-on light **1002** on the cord **102**. The clip may be sized for an interference fit or to "snap on" to the cord **102**. The clip **1010** may comprise a polymer or another suitable material. It may also vary in size or length to provide proper positioning of the bulb **112** relative to the cord **102** and/or components of the repellent dispersion ports **118**. Position A illustrates a relatively short clip **1010** while positions B and C illustrate a longer clip **1012** (which may otherwise be functionally identical to clip **1010**).

Position C also illustrates a clip-on light **1002** in a disassembled configuration. Here the battery **1008** may be seen outside the housing **1006** and the bulb **112** detached from the housing **1006**. It should be understood that internal springs for secure contact with the battery **1008** and circuitry as is known in the art for operation of the bulb **112** by the battery **1008** are included. The bulb **112** may be physically affixed to the housing **1006** by any suitable mechanism such as a bayonet style attachment, a friction fit, an interference fit, a press fit, a screw in fit, etc.

As described above the systems described herein (e.g., **100**, **400**, **700**, **800**, **900**) may be installed such that they are somewhere above ground level to allow maximum effectiveness of the repellent. In a specific embodiment, these may be installed such that the repellent dispersion ports are a maximum of about 100 feet from the ground or support surface. In such case, each repellent dispersion port (e.g., **118**) provides approximately a 330 square foot zone of protection from the respective port **118** down to the ground or support surface. Under mild conditions (e.g., low wind or no wind) the area of protection has a roughly circular appearance viewed from above.

Referring now to FIG. 11A is an overhead view of a plurality of repellent zones arranged in first plan according to aspects of the present disclosure. FIGS. 11A-D are in the context of a house **1102** or other structure and a protected porch, patio, or other area **1104**. As shown in FIG. 11A a lighting and repellent dispersal system **100** containing three repellent dispersion ports **118** is provided in a V-shape over the area **1104**. By arranging the system **100** such that the repellent dispersion ports **118** are spaced somewhat evenly over the area **1104** circular zones of coverage **1106** cover substantially all of the area **1104**. There are some areas **1108** overlapped by two zones **1106** and one area **1110** overlapped by three zones **1106**.

FIG. 11B is an overhead view of a plurality of repellent zones **1106** arranged in second plan according to aspects of the present disclosure. Here the system **100** is installed such that the repellent dispersion ports **118** are arranged at or near the perimeter or outer edge of the area **1104**. There is less overlap in this configuration but each repellent dispersion port **118** may be effective within its own zone **1106** such that overlap is not needed. Additionally, the total area protected by the three repellent zones **1106** is larger and actually extends beyond the area **1104**. As illustrated, using three repellent dispersion ports **118** at the perimeter of the area

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1104 leaves a small area **1112** near the house **1102** that is relatively unprotected. However, the repellent zones **1106** that surround this area **1112** may provide a sufficient “wall of coverage” that pests do not enter the area **1112**.

Referring now to FIG. **11C** an overhead view of a plurality of repellent zones **1106** arranged in a third plan according to aspects of the present disclosure is shown. Here the system **100** is arranged in an X-pattern with four active repellent dispersion ports **118** arranged in a square offset from the outside of the area **1104**. Here it can be seen that there are no “dead zones” and the protected zones **1106** extend beyond the edge of the area **1104** in every direction. Some areas **1108** are overlapped by two zones **1106** while others **1110** are overlapped by three, and others **1114** by four. It can also be seen that there are two idle repellent dispersion ports **118** on left and right sides of the area **1104**. Thus, it should be appreciated that not all ports **118** may need to be active to provide desired repellent performance.

Referring now to FIG. **11D** is an overhead view of a plurality of repellent zones **1106** arranged in third plan according to aspects of the present disclosure is shown. FIG. **11D** illustrates another way of deploying the system **100** when only three dispersion ports **118** are provided (or active). In this application, the system **100** is installed in a “zig-zag” pattern over the area **1104** with the ports **118** spaced roughly in a line laterally across the area **1104**. The zones **1106** provide full protection over the area **1104** including some double protected zones **1108** and one triple protected zone **1110** near the center of the area **1104**.

It will be appreciated that any of the systems described herein (e.g., **100**, **400**, **700**, **800**, **900**) can be adapted to cover a wide variety of areas. The protected areas are not necessarily square in every case (as in the examples of FIGS. **11A-D**) but can have a wide variety of geometric shapes. Additionally, the areas protected may vary substantially in size but can be protected and lighted by inclusion of additional dispersion ports and/or use of multiple strings of dispersion ports, lights, and clip-on devices as disclosed herein.

Referring now to FIG. **12A** is a perspective view of an embodiment of a retrofit assembly **1200** according to aspects of the present disclosure is shown. FIG. **12B** is a side view of the retrofit assembly **1200**. The retrofit assembly **1200** may be utilized to provide repellent features to a light string previously outfitted only with light bulbs (e.g., incandescent, LED, etc). In another aspect, the retrofit assembly **1200** provides an additional mechanism for a power cord to supply power to a heating receptacle (e.g., receptacle **123** discussed above or a heating receptacle **1210** as shown in FIGS. **12A-12B**) by fitting to the power cord in a similar fashion as a light bulb.

The retrofit assembly **1200** may comprise a receiver housing **1202** having a socket connector **1204** on a top end and a power cord **1208** exiting on or near a bottom end. The power cord **1208** transfers electrical power from the receiver housing **1202** to the heating receptacle **1208**. The heating receptacle **1208** may be suspended from the receiver housing **1202** by the power cord **1208** in a spaced apart manner (e.g., as shown). In other embodiments, wiring between the receiver housing **1202** and the heating receptacle is completely internal such that the heating receptacle **1210** and the receiver housing are a unitary structure.

The socket connector **1204** may comprise an E26/27 or S19 type connector or another light socket connector as known in the art. Power drawn from a light bulb receptacle

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on a power cord or elsewhere may be selectively passed to the heating receptacle **1210** based on electronics within the receiver housing **1202**.

In some embodiments, the receptacle housing **1210** comprises an indicator ring **1206**, which may comprise an LED. The indicator ring **1206** may illuminate when power is being supplied at the socket connector **1204** and is being passed down to the heating receptacle **1210**.

Referring now to FIG. **13A** an exploded view of the heating receptacle **1210** is shown. FIG. **13B** is a cutaway view of a heating receptacle **1210**. It should be understood that the heating receptacle **1210** is exemplary. The retrofit assembly **1200** could also utilize heating receptacle **123** or similar, for example. Heating receptacle **1210** may comprise a top shell **1302** which may be conic or generally sloped (e.g., to shed water or for decorative purposes). The top shell **1302** may be secured to a base **1304** by one or more fasteners **1306**. Fasteners **1306** may comprise screws that pass through the top shell **1302** to be received in the base **1304**.

The power cord **1208** (not shown here) may pass into the heating receptacle **1210** and be retained by a bushing **1308**. The bushing **1308** may comprise a rubber component or be formed of another elastomeric substance to provide a secure grip on the power cord and providing weather resistance.

Inside the heating receptacle **1210** the power cord may split into separate positive and negative leads (not shown here) and may pass through a bracket **1310** before connecting to a heating element **1314**. The heating element **1314** may comprise a resistive heating element. It may be secured to the base **1304** via one or more fasteners **1316**.

The base **1304** may comprise an upper ventilated portion **1322** connecting to a lower cylindrical sleeve **1324**. A floor **1328** may be retained at the junction of the ventilated portion **1322** and the lower sleeve **1324**, and may be affixed to either or both. In another embodiment, each of the upper ventilated portion **1322** and the lower sleeve **1324** affix to the floor **1328**. One or more of the floor **1328**, the upper ventilated portion, and the lower sleeve may have receptacles **1330** affixed thereto for receiving fasteners **1306**.

The floor **1328** may define an aperture **1320** for admitting a wick into or near to the heating element **1314** for dispersal of repellents. When the heating receptacle **1210** is assembled, the ventilated portion **1322** surround the heating element **1314** to allow dispersal of repellent outside of the heating receptacle **1210**. The heating element may be affixed to the floor **1328** on or near aperture **1320**.

A collar **1318** may be fitted into the lower sleeve **1324**. The collar **1318** may be sized to fit into the lower sleeve **1324** and rotate freely. Tabs **1604** or other interference members inside the sleeve **1324** may prevent the collar **1318** from falling out, while still allowing for rotation of the collar. The collar **1318** may serve as the interface to a detachable repellent pod. To that end, collar **1318** may define a threaded fitting **1303**. The fitting **1303** may selectively attached to a threaded repellent pod, such as that shown in FIG. **17**. The repellent pod **1700** is substantially similar to the repellent pod **126** discussed above, but for the threaded fitting **1702** to fit unto the fitting **1303** of collar **1318**.

Since the collar **1318** is free to rotate in the sleeve **1324**, the remainder of the retrofit assembly **1200** cannot be used as a grip to attach and detach the repellent pod **1700**. An opening or window **1326** may be defined in the sleeve **1324** to allow contact or grip with the collar **1318**. Rotation of the repellent pod **1700** with respect to the collar **1318** (or vice versa) may then be achieved for attachment and detachment of the repellent pod **1700**. The collar **1318** may have a knurled, ridged, or otherwise textured circumferential sur-

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face **1334** to increase purchase on the collar **1318**. In some embodiments, more than one window **1326** may be provided in the sleeve **1324** (for example, two windows 180 degrees apart).

An aperture **1332** may be defined centrally through the collar **1318** over the fitting **1303** to allow the wick or **134** or top port **206** of repellent pod **1700** or similar into proximity of the heating element **1314**.

Referring now to FIG. **14** a side, partial cutaway view of one embodiment of a receiver housing for use with systems of the present disclosure is shown. The receiver housing **1202** may contain electronic components that allow power to be selectively supplied to the heating receptacle **1210** from the socket connector **1204**. A negative lead **1406** and positive lead **1408** can be seen passing through the receiver housing **1202** (e.g., before they are joined together in power cord **1208**). A microcontroller **1404** or other solid-state device may be wirelessly controlled to interrupt or pass power to the heating receptacle **1210**. Thus the microcontroller **1404** functions as a remote controlled and solid state switch. The microcontroller **1404** may also illuminate indicator ring **1206** (e.g., via lead **1410**) when power is being passed to the heating receptacle. The microcontroller **1404** can be any wireless capable chip or system that can receive simple on/off signals from a transmitter. It may contain an interpreted or separate antenna as is known in the art. Those of skill in the art should understand that the wiring in FIG. **14** (and other figures) is exemplary only and function and controls described herein could be implemented in a number of ways.

In other embodiments, the receiver housing **1202** is eliminated and the power cord **1208** may connect directly to the socket connector **1204**. Such embodiments may contain the internal components of receiver housing **1204** in the heating receptacle **1210**, or may lack the remote-control capability of those embodiments with the internal electronic components of the receiver housing **1204**.

The microcontroller **1404** may receive signals from a number of devices. For example, it may be controlled by and provide responses or indications to a phone app via Bluetooth, Wi-Fi, or the like. Nearfield or wideband communication protocols may also be utilized. In some cases, a dedicated wireless remote control may be used. FIG. **15** is perspective view of a one embodiment of a wireless remote control **1500** for use with systems of the present disclosure. FIG. **16** is a simplified cutaway view of the wireless remote control of FIG. **15**.

The remote **1500** may comprise an outer casing **1501** containing a circuit board **1603** with a power supply such as a button cell battery **1606**. A microcontroller or other silicon **1604** chip device may take input from an on button **1502** and an off button **1504**. In response, the microcontroller **1604** may send wireless commands to the retrofit assembly **1200** to turn on or off depending on the button pressed. The microcontroller **1604** may contain an internal antenna or may utilize a separate antenna on the circuit board **1603** or elsewhere. An indicator light **1506** may be visible outside the casing **1501** to confirm that signals are being sent from the remote **1500**.

The remote **1500** may communicate with the retrofit assembly **1200** via radiofrequency protocols, as described. However, it may also be configured to communicate via infrared (IR), as is known in the art. In such cases, the retrofit assembly **1200** would be similarly configured to receive IR commands, as is known in the art. It should also be understood that a single remote **1500** may operate a plurality of retrofit devices such as the retrofit device **1200**.

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Referring now to FIG. **18**, a retrofitted string light **1800** utilizing retrofit assemblies **1200** according to the present disclosure is shown. Unlike previous embodiments, a power cord **1802** does not have cord junctions fitted to permanently attached heating receptacles. Instead, the power cord **1802** has only cord junctions **106** providing transverse power cords **108** leading to light sockets **110** (and, in some cases, there is no transverse cord, the light socket attaching directly to the cord junction **106** and/or the cord **102**).

Retrofit assemblies **1200** may be attached to some or all of the light sockets **110** in place of bulbs **112** for example. As can be seen, it is not necessary that all bulbs **110** be replaced. It may be desirable to replace only some of the plurality of bulbs **110** that would be used with a string light to allow for both repellency and lighting from the same retrofitted string light. Retrofit repellent assemblies **1200** may be spaced apart on the string **1802** (which is only partially shown, and would typically have many more sockets **110**) such that overlapping zones of repellent coverage are provided (for example, as in FIG. **11**, discussed above).

It is to be understood that the terms “including”, “comprising”, “consisting” and grammatical variants thereof do not preclude the addition of one or more components, features, steps, or integers or groups thereof and that the terms are to be construed as specifying components, features, steps or integers.

If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

It is to be understood that where the claims or specification refer to “a” or “an” element, such reference is not be construed that there is only one of that element.

It is to be understood that where the specification states that a component, feature, structure, or characteristic “may”, “might”, “can” or “could” be included, that particular component, feature, structure, or characteristic is not required to be included.

Where applicable, although state diagrams, flow diagrams or both may be used to describe embodiments, the invention is not limited to those diagrams or to the corresponding descriptions. For example, flow need not move through each illustrated box or state, or in exactly the same order as illustrated and described.

Methods of the present invention may be implemented by performing or completing manually, automatically, or a combination thereof, selected steps or tasks.

The term “method” may refer to manners, means, techniques and procedures for accomplishing a given task including, but not limited to, those manners, means, techniques and procedures either known to, or readily developed from known manners, means, techniques and procedures by practitioners of the art to which the invention belongs.

The term “at least” followed by a number is used herein to denote the start of a range beginning with that number (which may be a range having an upper limit or no upper limit, depending on the variable being defined). For example, “at least 1” means 1 or more than 1. The term “at most” followed by a number is used herein to denote the end of a range ending with that number (which may be a range having 1 or 0 as its lower limit, or a range having no lower limit, depending upon the variable being defined). For example, “at most 4” means 4 or less than 4, and “at most 40%” means 40% or less than 40%.

When, in this document, a range is given as “(a first number) to (a second number)” or “(a first number)-(a second number)”, this means a range whose lower limit is the first number and whose upper limit is the second number.

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For example, 25 to 100 should be interpreted to mean a range whose lower limit is 25 and whose upper limit is 100. Additionally, it should be noted that where a range is given, every possible subrange or interval within that range is also specifically intended unless the context indicates to the contrary. For example, if the specification indicates a range of 25 to 100 such range is also intended to include subranges such as 26-100, 27-100, etc., 25-99, 25-98, etc., as well as any other possible combination of lower and upper values within the stated range, e.g., 33-47, 60-97, 41-45, 28-96, etc. Note that integer range values have been used in this paragraph for purposes of illustration only and decimal and fractional values (e.g., 46.7-91.3) should also be understood to be intended as possible subrange endpoints unless specifically excluded.

It should be noted that where reference is made herein to a method comprising two or more defined steps, the defined steps can be carried out in any order or simultaneously (except where context excludes that possibility), and the method can also include one or more other steps which are carried out before any of the defined steps, between two of the defined steps, or after all of the defined steps (except where context excludes that possibility).

Further, it should be noted that terms of approximation (e.g., “about”, “substantially”, “approximately”, etc.) are to be interpreted according to their ordinary and customary meanings as used in the associated art unless indicated otherwise herein. Absent a specific definition within this disclosure, and absent ordinary and customary usage in the associated art, such terms should be interpreted to be plus or minus 10% of the base value.

The term “selective” or “selectively,” unless otherwise indicated, is taken to mean that the operation or function is capable of being performed by the structure or device in reference, but the operation or function may not occur continuously or without interruption. Furthermore, a selective or selectively performed operation may be one that the user or operator of a device or method may choose whether or when to perform, but the function or operation is nevertheless fully operative on or within the relevant device, machine, or method and the same includes the necessary structure or components to perform such operation.

Thus, the present invention is well adapted to carry out the objects and attain the ends and advantages mentioned above as well as those inherent therein. While the inventive device

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has been described and illustrated herein by reference to certain preferred embodiments in relation to the drawings attached thereto, various changes and further modifications, apart from those shown or suggested herein, may be made therein by those of ordinary skill in the art, without departing from the spirit of the inventive concept the scope of which is to be determined by the following claims.

What is claimed is:

1. A repellent assembly comprising:

- a receiver housing with a light socket connector;
- a heating receptacle electrically connected to the receiver housing, the heating receptacle having an internal electric heater and a fitting for selectively receiving a repellent pod assembly in a position proximate the electric heater to heat the repellent pod assembly to disperse repellent; and
- a remote-controlled switch in the receiver housing that switches power from the light socket connector to the heating receptacle.

2. The repellent assembly of claim 1, wherein the heating receptacle further comprises:

- a body defining a sleeve; and
- a collar rotatably fitted into the sleeve and defining the fitting for selectively receiving the repellent pod assembly in the position proximate the electric heater to heat the repellent pod assembly to disperse repellent.

3. The repellent assembly of claim 2, further comprising a window defined in the sleeve allowing user contact with the collar.

4. The repellent assembly of claim 3, wherein the collar has a textured outer circumferential surface contactable through the window.

5. The repellent assembly of claim 1, further comprising a length of power cord connecting the heating receptacle to the receiver housing.

6. The repellent assembly of claim 1, wherein the remote-controlled switch comprises a solid state switch.

7. The repellent assembly of claim 6, wherein the solid-state switch comprises a wireless communications capable microcontroller.

8. The repellent assembly of claim 7, further comprising a remote control provide communication to the microcontroller to power on and power off the heating receptacle.

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